

① Design a class hierarchy (pseudocode):

class person

 Data Members:

 Name

 Age

 Method:

 Input Person ()

 Display Person ()

End Class

Class Student inherits Person

 Data Members:

 Register Number

 Marks[5]

 Method:

 Calculate Average ()

 Sum = 0

 For i = 0 to 4

 Sum = Sum + Marks[i]

 End For

 Average = Sum / 5

 Return Average

End Class

Class Research Student inherits Student

 Data Members:

 Research Area

Method:

• DisplayCompleteDetails ()

Display Name

Display Age

Display RegisterNumber

Display Marks

Display Average

Display ResearchArea.

End class

Main program

create object Rs of ResearchStudent

Input Name

Input Age

Input RegisterNumber

For i = 0 to 4

Input Marks [i]

End for

input ResearchArea

Average = Rs. calculate Average ()

Rs. DisplayCompleteDetails ()

End Program.

②

Complete the Missing pseudocode.

BEGIN

READ N

READ ARRAY A

FOR $i \leftarrow 0$ TO $N-1$

FOR $j \leftarrow i+1$ TO $N-1$

IF $A[i] == A[j]$

$A[j] \leftarrow -1$

ENDIF

ENDFOR

ENDFOR

PRINT Modified Array

END

1. fill the missing logic.

1. FOR $j \leftarrow i+1$ TO $N-1$

2. $A[j] \leftarrow -1$.

③

Debug the Pseudocode.

Pseudocode :

Class Area

Method area (length)

RETURN length * length

END METHOD

METHOD area (length, breadth).

RETURN length * breadth

END METHOD

END CLASS.

Incorrect Line:

RETURN length * breadth .

Correct line:

RETURN length * length .

④

Trace the Execution:

Initial: Sum = 0

i	Even/Odd	operation	sum
1	odd	$0 - 1$	-1
2	Even	$-1 + 2$	1
3	odd	$1 - 3$	-2
4	Even	$-2 + 4$	2
5	odd	$2 - 5$	-3

Finally:

PRINT Sum

output: -3.

⑤ write Efficient Pseudocode:

BEGIN

READ N

READ A [0 ... N-1]

largest $\leftarrow -\infty$

second largest $\leftarrow -\infty$

FOR $i \leftarrow 0$ TO $N-1$

IF $A[i] > \text{largest}$ THEN

 second largest $\leftarrow \text{largest}$

 largest $\leftarrow A[i]$

ELSE IF $A[i] \geq \text{second largest}$ AND $A[i] < \text{largest}$ THEN

 second largest $\leftarrow A[i]$

ENDIF

ENDFOR

PRINT second largest

END.

⑥ Convert Problem Statement into Pseudocode.

```
INTERFACE Payment
    METHOD Pay (amount)
END INTERFACE

CLASS CreditCard IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Payment of ", amount, " using credit card"
    END METHOD
END CLASS

CLASS UPI IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Payment of ", amount, " using UPI"
    END METHOD
END CLASS

CLASS NetBanking IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Payment of ", amount, " using Net
            Banking"
    END METHOD
END CLASS
```

BEGIN

READ choice

READ amount

IF choice = 1 THEN

P ← credit Card

ELSE IF choice = 2 THEN

P ← UPI

ELSE

P ← Net Banking

ENDIF

P. Pay (amount)

END.

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Complete the Pseudocode.

pseudocode;

CLASS Animal

METHOD Speak()

PRINT "Animal"

END METHOD

END CLASS

CLASS Dog inherits Animal

METHOD Speak()

PRINT "Barks"

END METHOD

END CLASS

Animal obj ← new Dog()

obj.Speak()

-Answer:

* Bank 1: inherits

* Bank 2: new Dog()

⑧ Find Logical Errors

```
for id ← id  
    name ← name } . does not initialize the object.
```

Correct Constructor:

```
CONSTRUCTOR (id, name)
```

```
    this.id ← id
```

```
    this.name ← name
```

```
END CONSTRUCTOR.
```

- * Constructor should accept parameters id and name
- * we use this.id and this.name to assign the values to the object's data members.

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Case Study (integrated Problem).

CLASS Vehicle

DATA

VehicleNo

ownerName

METHOD display()

PRINT VehicleNo

PRINT ownerName

END METHOD

END CLASS

CLASS Car inherits Vehicle

DATA

FuelType

METHOD display()

PRINT VehicleNo

PRINT ownerName

PRINT FuelType

END METHOD

END CLASS

CLASS Truck inherits Vehicle

DATA

```

    loadCapacity
METHOD display()
    PRINT VehicleNo
    PRINT ownerName
    PRINT loadCapacity
END METHOD

END CLASS

BEGIN
    READ N
    FOR i ← 1 TO N
        READ type
        IF type = "Car" THEN
            obj ← new Car()
            READ obj.VehicleNo.
            READ obj.ownerName
            READ obj.fuelType
        ELSE
            obj ← new Truck()
            READ obj.VehicleNo
            READ obj.ownerName
            READ obj.loadCapacity
        ENDIF
        Store obj
    ENDFOR
    FOR EACH obj IN VehicleList obj.display()
    ENDFOR
END.

```